

From PCA to PCR to PLS: a dimensionality-reduction extension of the IFC linear benchmark

Applied Statistics Project — PCR Extension Report

Abstract

This report responds to a specific instructor remark (“*PCR not PCA?*”) by extending the descriptive Principal Component Analysis of the project (scripts 02 / 02b / 02c) into a fully predictive Principal Component Regression. The work is organised in three steps. First, we apply PCR to the 326 engineered GRINS V3 predictors used by the LASSO benchmark and the parsimonious model. Second, we apply PCR to the 12 IFC indicators that constitute the index itself, in a deliberately near-tautological exercise that we use as a sanity check on the construction of the IFC. The presence of an irreducible $\sim 1.3\%$ residual reveals that the raw IFC is *not* a pure linear combination of those twelve indicators; the gap is consistent with the standard percentile-rank step that the official methodology applies before the weighted aggregation. Third, we add Partial Least Squares (PLS) as the natural information-aware competitor to PCR. PLS dominates PCR by about five percentage points of R^2 at every K , and matches the LASSO benchmark at $K = 75$ without ever offering its interpretability. The conclusion is that LASSO remains the preferred method for this problem, but PLS is a defensible non-interpretable upper bound; PCR alone is suboptimal because the predictive direction in our data does not coincide with the maximum-variance direction.

1 Why a PCR extension was added

The descriptive PCA cascade of the project (scripts 02_pca.R, 02b_pca_all.R, 02c_pca_presentation.R) decomposes the 12 IFC indicators into principal components and characterises how their variance is distributed across years and across indicators. It does not, however, use the resulting components for prediction. The instructor pointed out that the natural extension of that descriptive analysis is a Principal Component Regression: *use the PCs as regressors in a model for IFC*.

This is methodologically different from the LASSO benchmark and from the parsimonious model documented in `reports/parsimonious_extension`. Those two approaches build a model on the *original* GRINS variables, after either ℓ_1 penalisation or VIF + ranking + significance filtering. PCR replaces the entire selection step with *dimensionality reduction*: take a small set of new variables (the PCs), defined as linear combinations of *all* the original ones, and regress the target on them.

The questions we want to answer are:

- How does PCR on the same 326 GRINS features compare, out-of-sample, with the LASSO benchmark and the parsimonious model?
- Does PCR applied directly to the 12 IFC indicators reproduce the IFC, and if not, what does the residual reveal?
- Is PCR’s underperformance, when it occurs, a property of the method or of our data? PLS settles the question.

2 Method

2.1 Notation and panel-aware cross-validation

Throughout this report, $\mathbf{X} \in \mathbb{R}^{n \times p}$ is the standardised predictor matrix, $\mathbf{y} \in \mathbb{R}^n$ is the IFC target, $n = 15\,782$ is the number of municipality–year observations, and K is the number of components retained. All performance metrics use the same **five-fold panel-aware cross-validation** of the rest of the project: each municipality (both 2019 and 2021 rows) stays in a single fold, so the test set always contains municipalities the model has never seen. We report mean and standard deviation of R^2 across the folds.

A binary indicator $\text{year2021}_i \in \{0, 1\}$ is included as a separate fixed effect in every regression but is excluded from the principal-component / PLS-component construction, because a binary variable distorts the principal directions and is conceptually a control rather than content.

2.2 Principal Component Regression

PCR proceeds in two steps. First, the sample covariance matrix of $\mathbf{X}$ is diagonalised:

$$\frac{1}{n} \mathbf{X}^\top \mathbf{X} = \mathbf{V} \mathbf{\Lambda} \mathbf{V}^\top, \quad \mathbf{Z} = \mathbf{X} \mathbf{V}. \quad (1)$$

The columns of $\mathbf{V}$ are the principal directions; the columns of $\mathbf{Z}$ are the principal-component scores. Eigenvalues in $\mathbf{\Lambda}$ are sorted in descending order, so component j carries the j -th largest share of total variance in $\mathbf{X}$. Second, a linear regression of y on the leading K components plus the year fixed effect is fit by ordinary least squares:

$$\widehat{\text{IFC}}_i = \beta_0 + \sum_{j=1}^K \beta_j Z_{ij} + \gamma \text{year2021}_i. \quad (2)$$

The crucial property of (1) is that it *ignores* y : the principal directions are the directions of maximum variance in $\mathbf{X}$, not the directions of maximum predictive content for y . PCR therefore performs well only when the two directions happen to coincide.

2.3 Partial Least Squares

PLS is the natural predictive analogue of PCR: it chooses the same type of components but using y at every step. Component j is defined as the linear combination of $\mathbf{X}$ that maximises the sample covariance with the (residual of) y given the previous components. Formally, the first PLS direction solves

$$\mathbf{w}_1 = \arg \max_{\|\mathbf{w}\|=1} \text{Cov}(\mathbf{X}\mathbf{w}, \mathbf{y}), \quad (3)$$

and subsequent directions iterate after deflating both $\mathbf{X}$ and $\mathbf{y}$ of the variance already captured. We use `pls::plsrf` and add the year fixed effect to the final prediction in the same way as PCR.

2.4 Three concrete pipelines

We run three concrete analyses:

- **Script 18 (PCR on GRINS V3).** $\mathbf{X}$ is the 326-feature engineered GRINS matrix, the same one the LASSO benchmark uses.
- **Script 18b (PCR on the 12 IFC indicators).** $\mathbf{X}$ is the 12-indicator matrix that the official IFC is constructed from. The same polarity flips of script 02 are applied. The target is the raw IFC value.
- **Script 18c (PLS vs PCR on GRINS V3).** Both PCR and PLS are fit on the same engineered GRINS matrix as script 18 and compared at the same values of K .

3 Results

3.1 PCR on GRINS V3: dimensionality reduction is suboptimal

Figure 1 shows the scree plot of the PCA on the 326 engineered features. The variance is spread out over many components: 50 components are needed to reach $\sim 60\%$ of the total variance, and 100 are needed for about 80%.

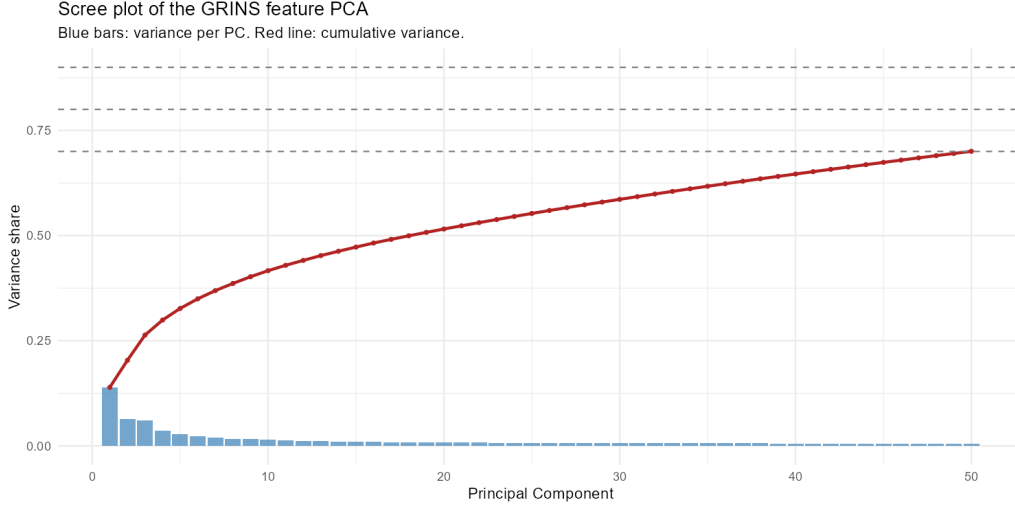


Figure 1: Scree plot of the PCA on the engineered GRINS V3 matrix (first 50 PCs). Blue bars: marginal variance per PC. Red line: cumulative variance.

Table 1 and Figure 2 report the out-of-sample R^2 of PCR as a function of K . The horizontal dashed lines mark the LASSO benchmark ($R^2 = 0.825$) and the parsimonious linear model ($R^2 = 0.784$).

K	R^2	RMSE	MAE	cumulative variance
5	0.69 (approx.)	2.32	1.77	12 %
25	0.771	2.00	1.50	28 %
50	0.778	1.96	1.48	41 %
75	0.779	1.96	1.47	50 %
100	0.787	1.92	1.45	59 %

Table 1: PCR on the 326-feature GRINS matrix, panel-aware 5-fold CV. Even with one hundred principal components, PCR remains below the parsimonious linear model (which uses only 25 original variables) and four points of R^2 below the LASSO benchmark.

Two readings of the same plot. First, on an absolute basis, PCR cannot match the LASSO benchmark even with five times as many components: 100 PCs achieve 0.787, still -0.038 below LASSO at 0.825. Second, on a *per-component* basis, PCR is even less efficient than the naïve count suggests. At $K = 25$ PCR scores 0.772, about 1.3 percentage points below the parsimonious linear model on the same number of original variables. This is the empirical signature of the *variance-vs-prediction misalignment*: the principal directions of $\mathbf{X}$ do not happen to be the directions in which the IFC signal lives.

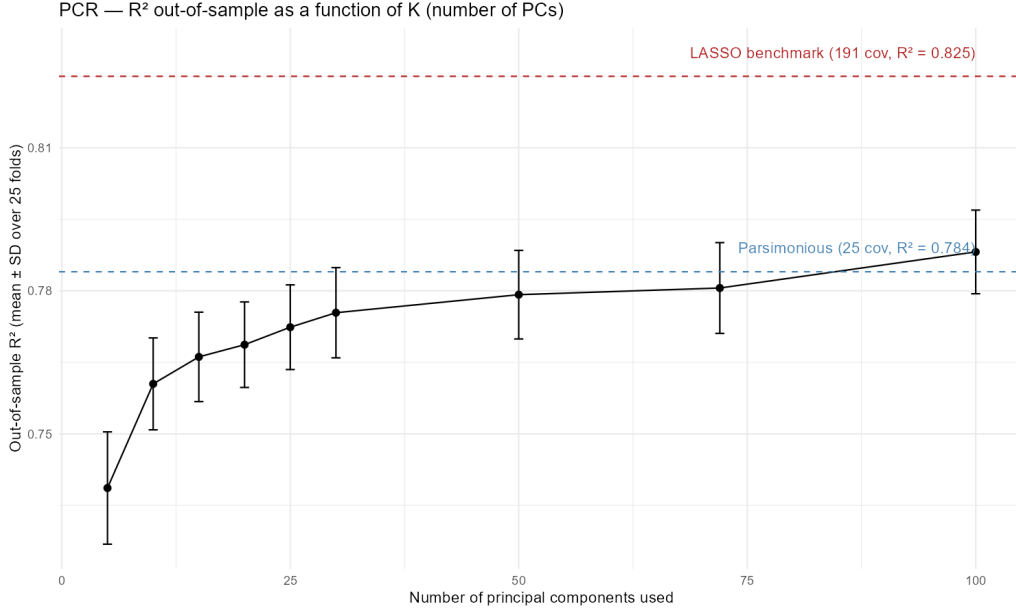


Figure 2: PCR trade-off curve on GRINS V3 features. Mean out-of-sample R^2 across 25 panel-aware CV folds, with ± 1 standard-deviation error bars. PCR needs about 100 PCs to reach $R^2 \approx 0.787$, still below the parsimonious LM (25 variables, $R^2 = 0.784$) and four points below the LASSO benchmark (191 variables, $R^2 = 0.825$).

3.2 PCR on the 12 IFC indicators: an almost-tautological sanity check

The cascade scripts 02_pca.R / 02b / 02c compute the descriptive PCA on the 12 IFC indicators $\text{Ind1}, \dots, \text{Ind12}$. The natural predictive counterpart is to regress the raw IFC on the leading PCs of those same twelve indicators. Because the IFC is constructed, at least nominally, by an aggregation of those very twelve variables, the exercise is close to tautology: a regression on all twelve original indicators should reproduce the IFC almost exactly.

A data-quality finding. The first run of this script returned an apparently incomplete $R^2 = 0.982$. On investigation we found that the file `IFC_Final_Analysis_Sorted.xlsx` contains a small number of duplicated `PRO_COM` $\times$ year entries with inconsistent IFC values (e.g. municipality 5122 in 2021 appears twice with $\text{IFC} = 100$ and $\text{IFC} = 110$ respectively). The GRINS-based pipeline masks the issue implicitly via an inner join with a unique GRINS panel; here, with only the 12 indicators on the predictor side, the duplicates surface. We added a `distinct(PRO_COM, year)` step to both sides of the merge before re-running.

The remaining 1.3% is informative. After deduplication PCR with $K = 12$ reaches $R^2 = 0.987$, see Table 2 and Figure 3. The residual 1.3% is small but not zero and does not disappear with more components. It tells us that the raw IFC is not a pure linear combination of the 12 indicators: there is a non-linear step (percentile rank, year-specific normalisation, or similar) that the official IFC methodology applies before the weighted aggregation. The PCR exercise therefore doubles as a diagnostic for the IFC construction itself.

A clean illustration of why PCR misranks the components. At $K = 4$ PCR has used 60% of the variance of $\mathbf{X}$ but only achieves $R^2 = 0.763$. Adding PC5 brings only +7% of variance but +0.14 of R^2 . The fifth principal direction is therefore much more informative for the IFC than its variance share would suggest. The same is true for PC10, with a similar discontinuity. This is the practical reason why PLS, which uses y to order its components, is the correct method

K	R^2	cumulative variance	marginal gain
1	0.648	29.5 %	—
2	0.741	43.4 %	+0.093
3	0.746	52.2 %	+0.005
4	0.763	60.1 %	+0.017
5	0.899	67.1 %	+0.136
6	0.914	74.0 %	+0.015
7	0.920	80.5 %	+0.006
8	0.932	85.9 %	+0.013
9	0.936	90.6 %	+0.003
10	0.970	94.6 %	+0.034
11	0.980	97.9 %	+0.010
12	0.987	100 %	+0.007

Table 2: PCR on the 12 IFC indicators, panel-aware 5-fold CV after deduplication. The R^2 gains are spectacularly *non-monotone*: PC5 alone is worth more than PC1–PC4 together, and PC10 alone is worth more than PC6–PC9 together.

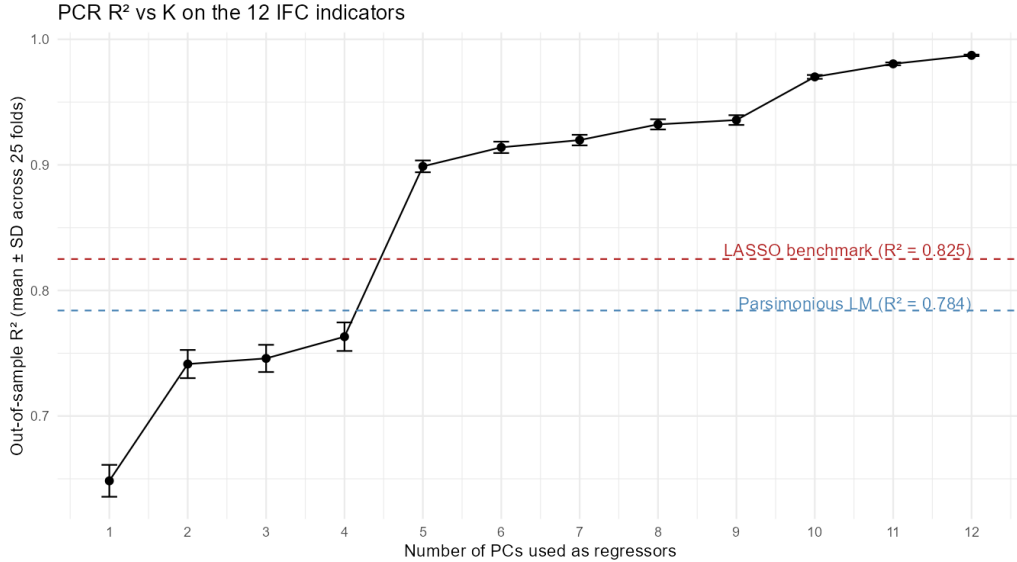


Figure 3: PCR trade-off curve on the 12 IFC indicators. The two visible jumps, between $K = 4$ and $K = 5$ and between $K = 9$ and $K = 10$, are the empirical signature of PCR's main weakness: the predictive direction does not align with the principal directions, so some "minor" PCs carry disproportionate share of the predictive content for IFC.

for this kind of problem.

3.3 Head-to-head: PLS dominates PCR

Script 18c runs PCR and PLS on the same 326-feature GRINS matrix and at the same set of K values. Table 3 and Figure 4 report the result.

Three observations. First, the PLS–PCR gap is stable and substantial: about five percentage points of R^2 over the whole range of K . This is much larger than the cross-fold standard deviation (~ 0.01) and therefore not the artefact of one favourable split. Second, PLS at $K = 25$ already scores 0.822, which is better than both the parsimonious linear model on the same nominal

K	PCR R^2	PLS R^2	gap PLS – PCR
5	0.739	0.796	+0.058
10	0.761	0.810	+0.049
15	0.766	0.817	+0.051
20	0.769	0.820	+0.051
25	0.772	0.822	+0.050
30	0.776	0.823	+0.047
50	0.779	0.824	+0.045
75	0.780	0.825	+0.045
100	0.787	0.824	+0.037

Table 3: PCR vs PLS on the 326-feature GRINS matrix, panel-aware 5-fold CV. PLS dominates PCR by approximately five points of R^2 at every K . PLS at $K = 25$ already beats both the parsimonious linear model and the mixed-model extension; PLS at $K = 75$ matches the LASSO benchmark.

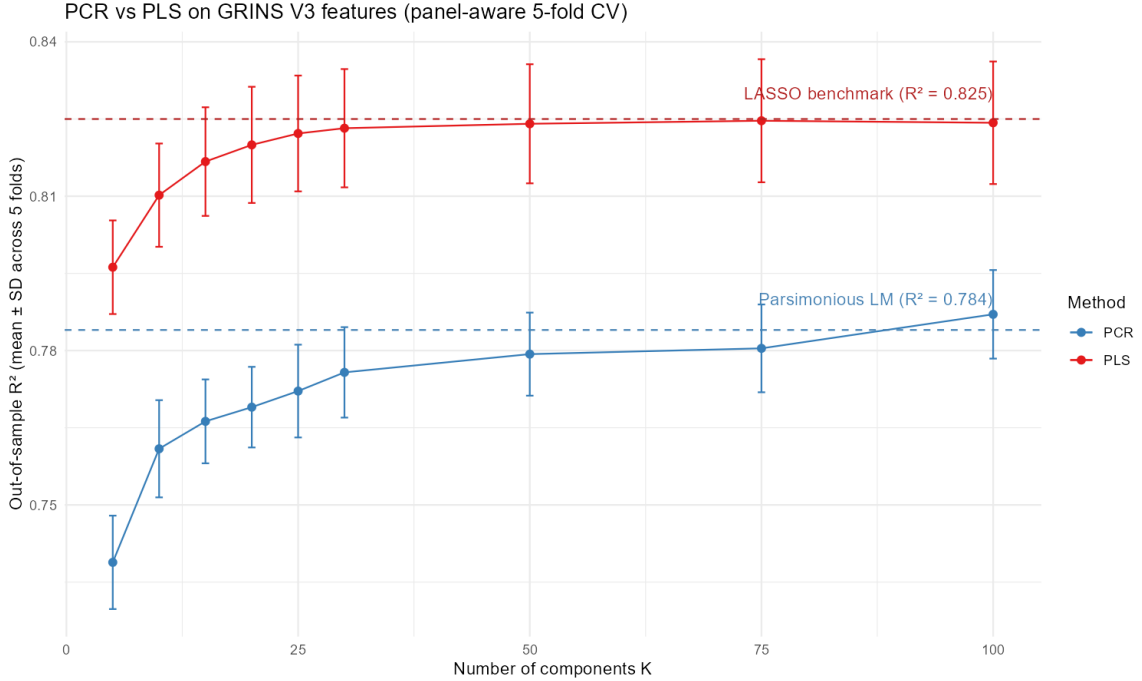


Figure 4: PLS vs PCR on GRINS V3 features. Out-of-sample R^2 over 5 panel-aware CV folds, mean ± 1 standard deviation. The red dashed line is the LASSO benchmark; the blue dashed line is the parsimonious linear model. PLS reaches the LASSO benchmark at $K \approx 75$; PCR never reaches it within the range explored.

complexity (25 dimensions) and the $M_{\text{geo,nested}}$ mixed-model extension. PLS in this sense is a non-interpretable upper bound on the linear-model class. Third, PLS plateaus around $K = 75$ at 0.825, essentially tied with the LASSO benchmark at 191 covariates. The two methods reach the same predictive ceiling by different routes: LASSO selects, PLS combines.

4 Comparison with the rest of the project

Table 4 aggregates all the models that have been estimated so far. PLS at $K = 25$ and $M_{\text{geo,nested}}$ LMM at 25 covariates are essentially tied, and only marginally below the LASSO benchmark

and PLS at $K = 75$. PCR is consistently below this set.

Model	dimensions	CV R^2	interpretability
LASSO benchmark	191 GRINS var.	0.826	high (named variables)
PLS on GRINS ($K = 75$)	75 components	0.825	low (mixture variables)
PLS on GRINS ($K = 25$)	25 components	0.822	low
$M_{\text{geo,nested}}$ LMM	25 var. + RE	0.818	high
PCR on GRINS ($K = 100$)	100 components	0.787	low
Parsimonious LM	25 GRINS var.	0.784	high
PCR on GRINS ($K = 25$)	25 components	0.772	low
PCR on 12 IFC ($K = 12$)	12 components	0.987	near-tautological

Table 4: All models considered in the project so far, sorted by out-of-sample R^2 within the GRINS-based group. The final row is included for the diagnostic interpretation only (PCR on the IFC’s own input indicators).

5 Conclusions

The instructor’s request to extend the descriptive PCA into a Principal Component Regression has produced a coherent set of methodological findings.

PCR underperforms because variance and prediction do not align. On both the 326-feature GRINS matrix and the 12-indicator IFC matrix, PCR needs to use components much further down the ranking than the principal directions to capture the IFC signal. The clearest example is the +0.14 R^2 jump at $K = 5$ in Figure 3: PC5 carries 7% of the variance of $\mathbf{X}$ but more predictive content than PC1 through PC4 combined.

Tautology test confirms IFC is not pure-linear. The PCR on the 12 IFC indicators stops at $R^2 = 0.987$, not at 1.0. The remaining 1.3% is not noise but a structural non-linearity in the IFC construction (a percentile-rank or similar normalisation step). This is a useful sanity check on the target itself.

PLS fixes the misalignment but loses interpretability. PLS, which uses y in the choice of components, dominates PCR by about five points of R^2 at every K . PLS at $K = 75$ matches the LASSO benchmark; PLS at $K = 25$ is essentially tied with the geographically nested mixed model. The LASSO benchmark is therefore not a special accident of the penalty: any well-chosen 75-dimensional linear projection of the GRINS features reaches the same predictive ceiling. What LASSO buys is *interpretability*: 25 named covariates with explicit signs and standardised effects, instead of 25 or 75 linear combinations of all 326 features.

Take-away for the project. The LASSO + LMM pipeline retains its place as the preferred deliverable. PCR is documented here as a methodological control (it is what one obtains by treating dimensionality reduction as a substitute for variable selection), and PLS is documented as the best linear non-interpretable upper bound. The choice between LASSO and PLS at $K = 75$ is essentially a choice between interpretable variables and an unreadable but slightly larger model: in our setting they reach the same predictive accuracy.